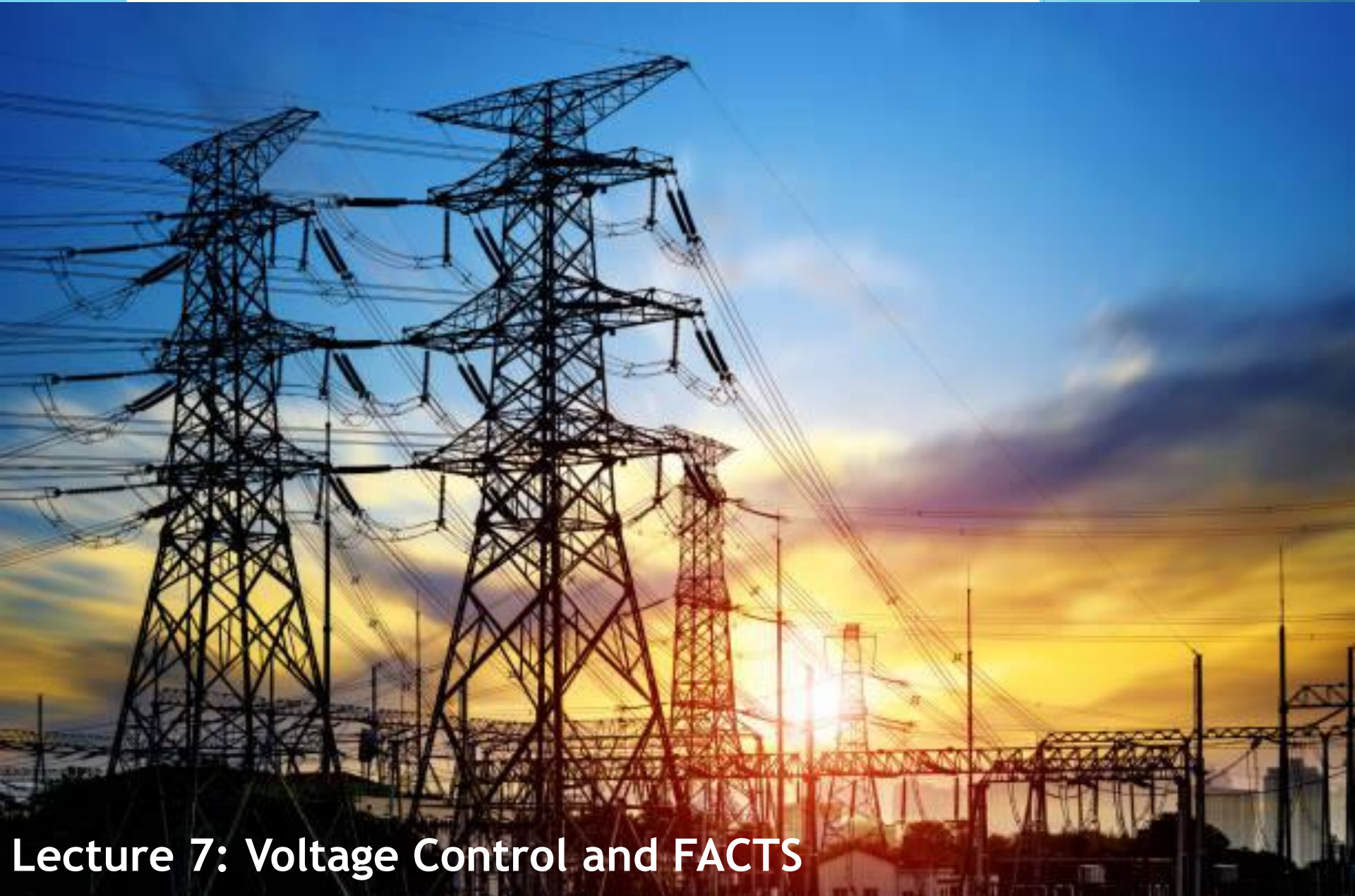




Lecture 7: Voltage Control and FACTS

EEE 321: Power System I

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Presentation Outline

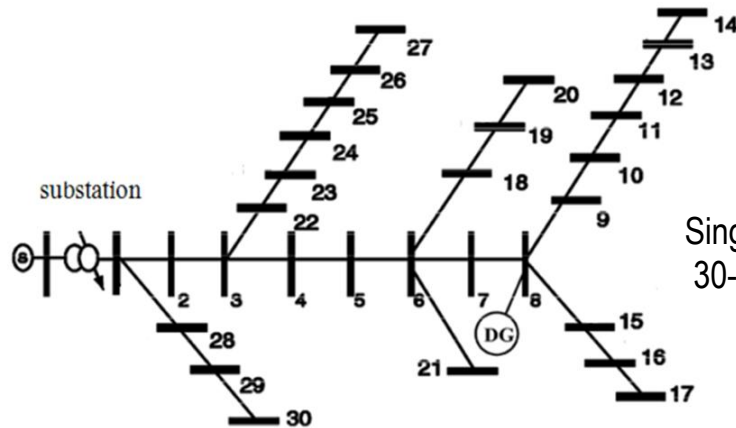
- ❑ Importance of Voltage Control
- ❑ Methods of Voltage Control
- ❑ Introduction to Flexible AC Transmission System (FACTS)
- ❑ Types of FACTS
- ❑ Applications of FACTS
- ❑ Advantages and Disadvantages of FACTS

IMPORTANCE OF VOLTAGE CONTROL

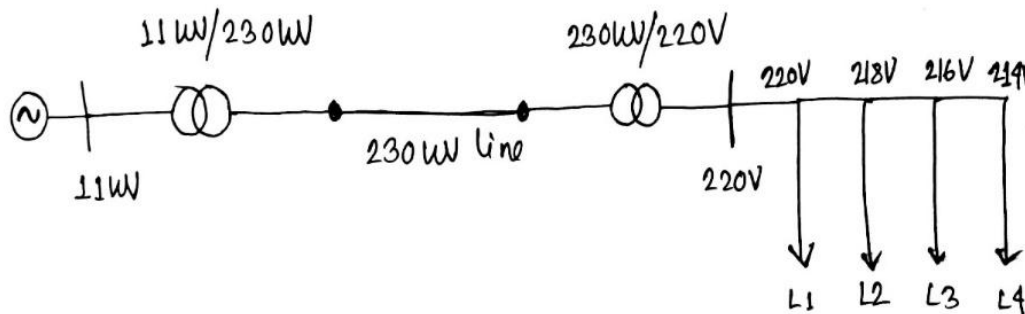
The **voltage at the consumer's terminal changes with change of load** on the supply system.

Example of One Feeder:

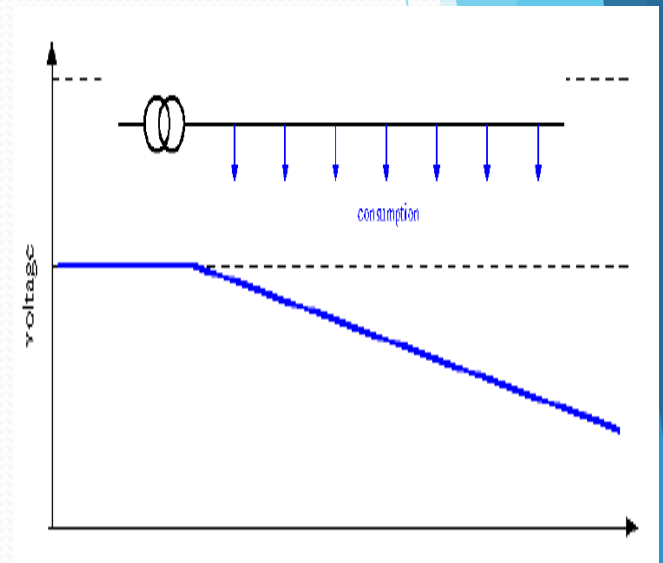
Voltage drop along the line increases with the increase in load.



Single line diagram of 30-bus IEEE radial test feeder



Allowable variation of voltage = $\pm 5\%$



IMPORTANCE OF VOLTAGE CONTROL

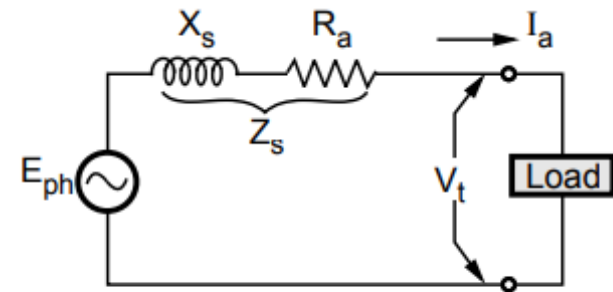
The variations of voltage at the consumer's terminals are undesirable. Low voltage in a distribution system can cause a variety of problems, primarily impacting the performance and lifespan of electrical equipment and appliances. It can lead to inefficient operation, reduced functionality, and even damage to devices. Additionally, it can result in increased energy consumption and potentially pose safety hazards.

- (1) In case of **lighting load**, the lamp characteristics are very sensitive to changes of voltage. For instance, if the supply voltage to an incandescent lamp decreases by 6% of the rated value, the illumination power may decrease by 20%. On the other hand, if the supply voltage is 6% above the rated value, the life of the lamp may be reduced by 50% due to rapid deterioration of the filament.
- (2) In case of **induction motor**, the voltage variation may cause erratic operation. If the supply voltage is above the normal, the motor may operate with a saturated magnetic circuit, with consequent large magnetizing current, heating and low power factor. On the other hand, if the voltage is too low, it will reduce the starting torque of the motor considerably. **When an induction motor experiences low voltage, it draws a larger current to compensate and maintain its required torque output.** The higher current drawn by the motor due to low voltage can cause it to overheat, potentially damaging the motor windings and insulation if the voltage drop is significant and sustained.
- (3) To **compensate for the low voltage, some devices may draw more current**, leading to higher energy consumption and increased electricity bills.

METHODS OF VOLTAGE CONTROL

There are several methods of voltage control. In each method, the system voltage is changed in accordance with the load to obtain a fairly constant voltage at the consumer's end of the system. The following are the methods of voltage control in an *a.c. power system:

- (i) By excitation control
- (ii) By using tap changing transformers
- (iii) Auto-transformer tap changing
- (iv) Booster transformers
- (v) Induction regulators
- (vi) By Reactive Power Compensator



Equivalent circuit of an alternator

BY EXCITATION CONTROL

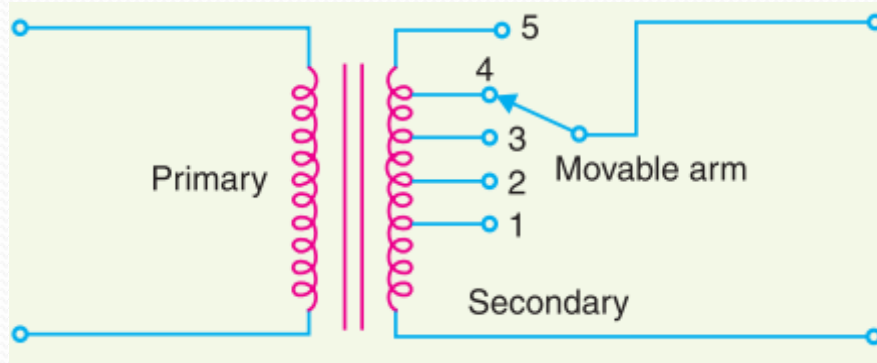
When the load on the supply system changes, the terminal voltage of the alternator also varies due to the changed voltage drop in the synchronous reactance of the armature. The voltage of the alternator can be kept constant by changing the *field current of the alternator in accordance with the load. This is known as *excitation control* method. The excitation of alternator can be controlled by the use of automatic or hand operated regulator acting in the field circuit of the alternator. The first method is preferred in modern practice. There are two main types of automatic voltage regulators viz.

- (i) Tirril Regulator
- (ii) Brown-Boveri Regulator

METHODS OF VOLTAGE CONTROL

BY USING TAP CHANGING TRANSFORMER

A number of tapings are provided on the secondary of transformer.



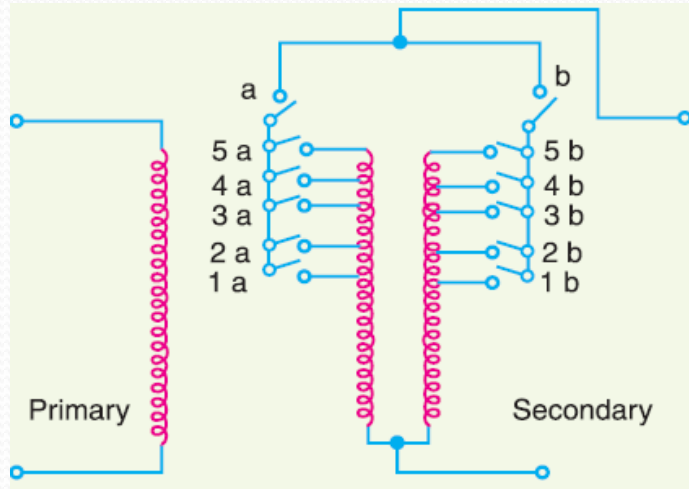
Off Load Tap-changing Transformer

- ☐ The effective number of secondary turns is varied.
- ☐ Secondary voltage is minimum when the movable arm makes contact with stud 1.
- ☐ Secondary voltage is maximum when the movable arm makes contact with stud 5.
- ☐ During the period of light load, the voltage across the primary is not much below the supply voltage and movable arm is placed on stud 1.
- ☐ When the load increases, the voltage across the primary drops, but the secondary voltage needs to be kept constant.
- ☐ The load is kept off during the changing of tap.

METHODS OF VOLTAGE CONTROL

BY USING TAP CHANGING TRANSFORMER

A number of tapings are provided on the secondary of transformer.



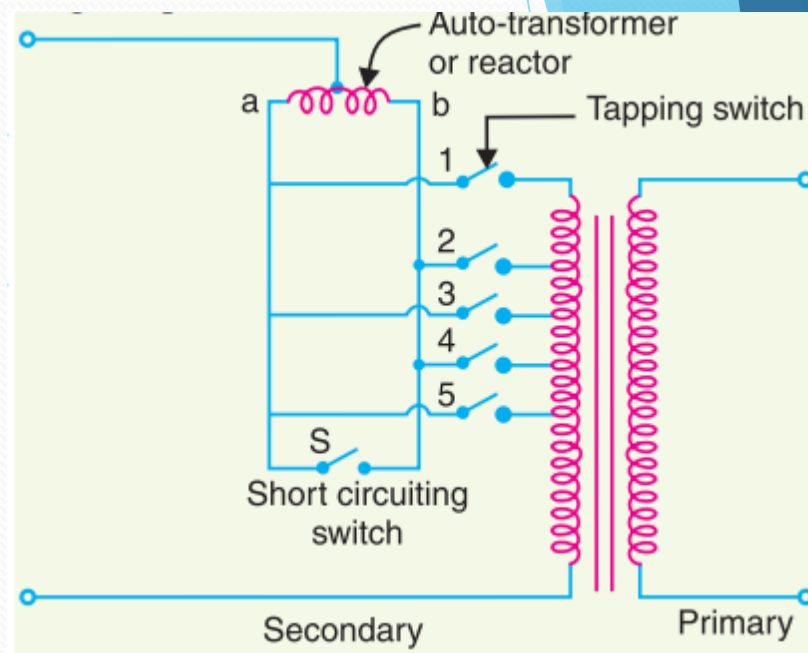
On Load Tap-changing Transformer

- ☐ No interruption to supplies.
- ☐ The impedance of the transformer is increased and there will be a voltage surge during switching.
- ☐ There are twice as many tapings as the voltage stage.

METHODS OF VOLTAGE CONTROL

BY AUTO-TRANSFORMER TAP CHANGING

Fig. 15.6 shows diagrammatically auto-transformer tap changing. Here, a mid-tapped auto-transformer or reactor is used. One of the lines is connected to its mid-tapping. One end, say a of this transformer is connected to a series of switches across the odd tapplings and the other end b is connected to switches across even tapplings. A short-circuiting switch S is connected across the auto-transformer and remains in the closed position under normal operation. In the normal operation, there is *no inductive voltage drop across the auto-transformer. Referring to Fig. 15.6, it is clear that with switch 5 closed, minimum



secondary turns are in the circuit and hence the output voltage will be the lowest. On the other hand, the output voltage will be maximum when switch 1 is closed.

Suppose now it is desired to alter the tapping point from position 5 to position 4 in order to raise the output voltage. For this purpose, short-circuiting switch S is opened, switch 4 is closed, then switch 5 is opened and finally short-circuiting switch is closed. In this way, tapping can be changed without interrupting the supply.

METHODS OF VOLTAGE CONTROL

BY BOOSTER TRANSFORMER TAP CHANGING

Sometimes it is desired to control the voltage of a transmission line at a point far away from the main transformer. This can be conveniently achieved by the use of a booster transformer as shown in Fig. 15.7. The secondary of the booster transformer is connected in series with the line whose voltage is to be controlled. The primary of this transformer is supplied from a regulating transformer *fitted with on-load tap-changing gear. The booster transformer is connected in such a way that its secondary injects a voltage in phase with the line voltage.

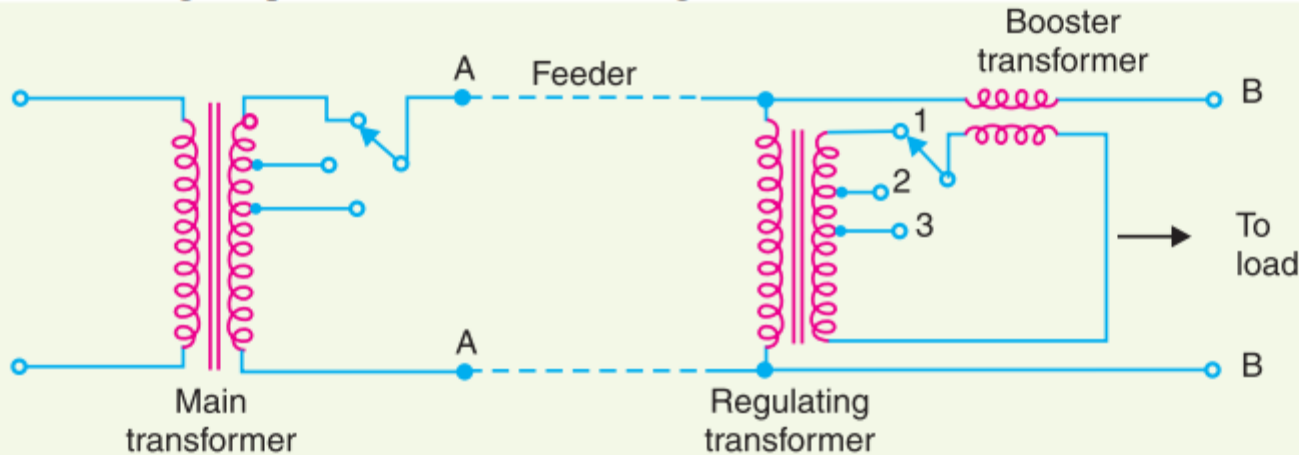


Fig. 15.7

METHODS OF VOLTAGE CONTROL

BY INDUCTION REGULATOR

An induction regulator is essentially a constant voltage transformer, one winding of which can be moved *w.r.t.* the other, thereby obtaining a variable secondary voltage. The primary winding is connected across the supply while the secondary winding is connected in series with the line whose voltage is to be controlled. When the position of one winding is changed *w.r.t.* the other, the secondary voltage injected into the line also changes. There are two types of induction regulators *viz.* single phase and 3-phase.

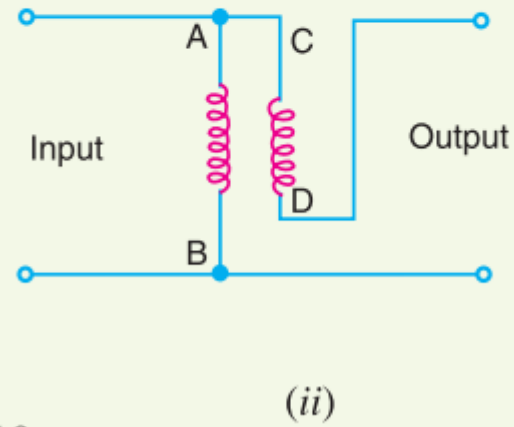
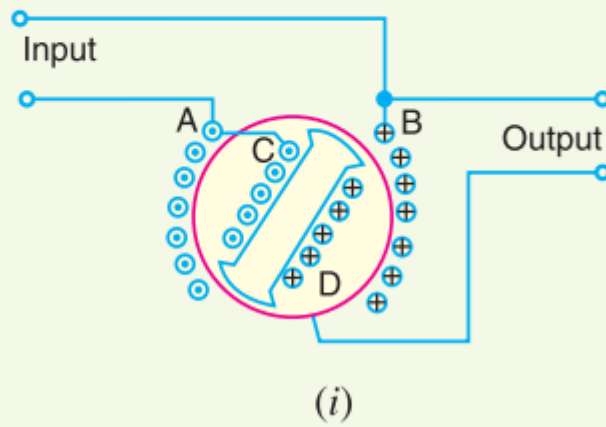


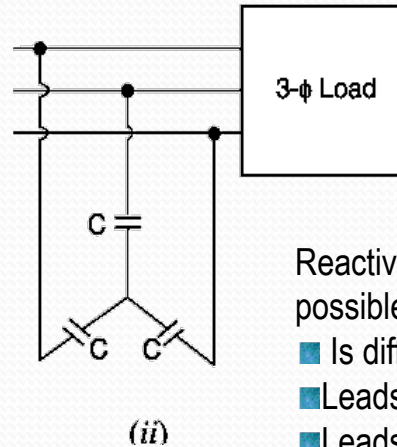
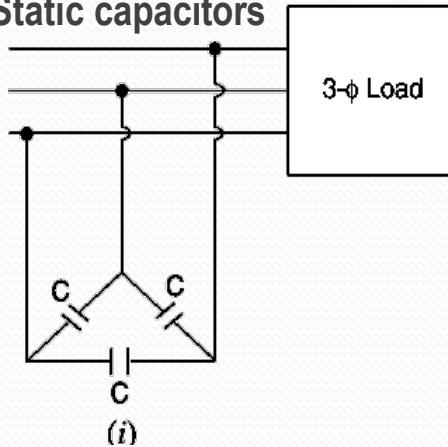
Fig. 15.9

METHODS OF VOLTAGE CONTROL

BY REACTIVE POWER COMPENSATOR

Reactive power compensators help balance reactive power demand, reduce energy losses, and improve the power factor.

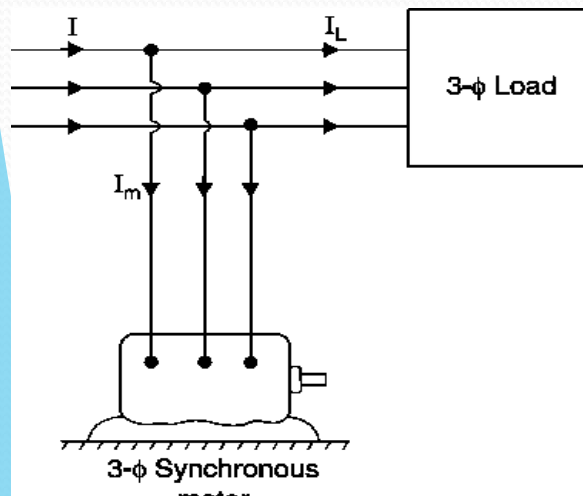
Static capacitors



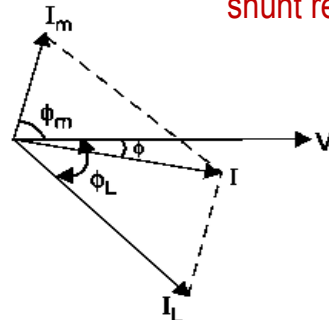
Reactive power compensation should be done as close as possible to the inductive loads, since reactive power:

- Is difficult to transport on the grid.
- Leads to extra losses on the lines.
- Leads to voltage drops on the lines, limiting their capacity.

Synchronous Condensers



Local generation or absorption of Q can be achieved by connecting shunt capacitors/synchronous condensers or shunt reactors (shunt inductors).



FLEXIBLE AC TRANSMISSION SYSTEM (FACTS)

FACTS stands for Flexible Alternating Current Transmission System or simply Flexible AC Transmission System. It is a power electronic based system where static devices are used to enhance and increase the power transfer capability and controllability. These devices are used in a power system network to enhance:

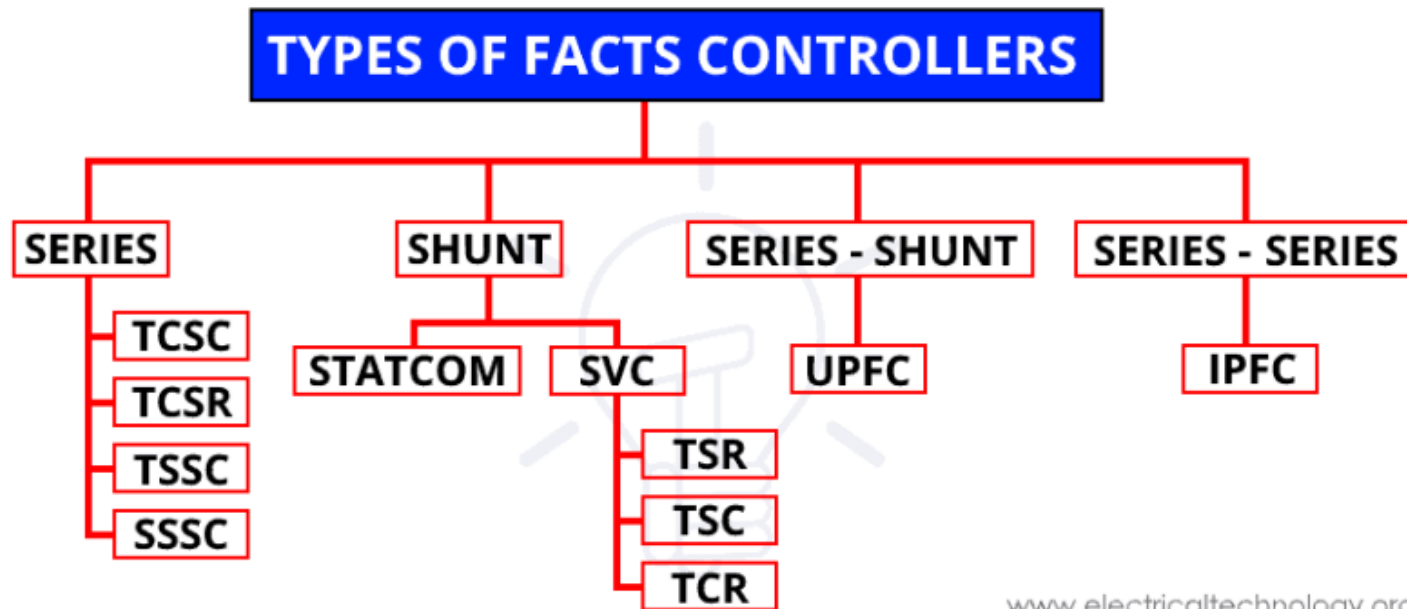
- Power System Stability,
- The System Utilization Capacity,
- Power flow control and power transfer capacity,
- Voltage regulation,
- Mitigation of system oscillations,
- System Security and Reliability
- Power quality of system interconnections.

Before the invention of power electronics switches, these problems were solved by connecting capacitor, reactor, or synchronous generator with the help of mechanical switches. But there is a lot of problems to use the mechanical switches. It has a very slow response and there is a problem of wear and tear of mechanical switches. So, these are not reliable ways to increase the controllability and stability of the transmission line.

FLEXIBLE AC TRANSMISSION SYSTEM (FACTS)

According to the type of connection FACTS Controller with the power system, it is classified as;

- ☐ Series Connected Controller
- ☐ Shunt Connected Controller
- ☐ Combined Series-Series Controller
- ☐ Combined Shunt-Series Controller



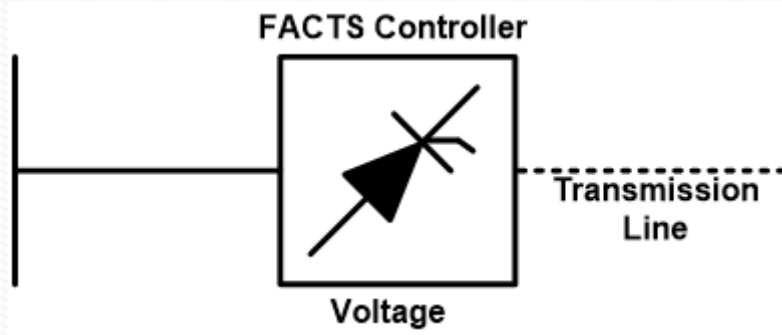
FLEXIBLE AC TRANSMISSION SYSTEM (FACTS)

Series Connected Controller

The series controllers are used to introduced voltage in series with the line voltage. It consists of a capacitor or reactor (impedance devices). This type of controllers is used to supply or consume variable reactive power.

When the load of the transmission line is more, it needs extra reactive power. In this condition, it is used to supply reactive power with the help of a capacitor. When the transmission line operates on very light load, the receiving end voltage increase than the sending end voltage due to less demand for reactive power. In this condition, it is used to consumes reactive power with the help of an inductor.

In most of the case, the capacitors are installed at the end of the line to compensate demand for reactive power. For this purpose, Thyristor Controlled Series Capacitor (TCSC), and Static Synchronous Series Compensator (SSSC) are used. The below figure shows the basic diagram of series connected Controller.



Thyristor Controlled Series Capacitor (TCSC)
Thyristor Controlled Series Reactor (TCSR)
Thyristor Switched Series Capacitor (TSSC)
Static Synchronous Series Compensator (SSSC)

FLEXIBLE AC TRANSMISSION SYSTEM (FACTS)

Series Connected Controller

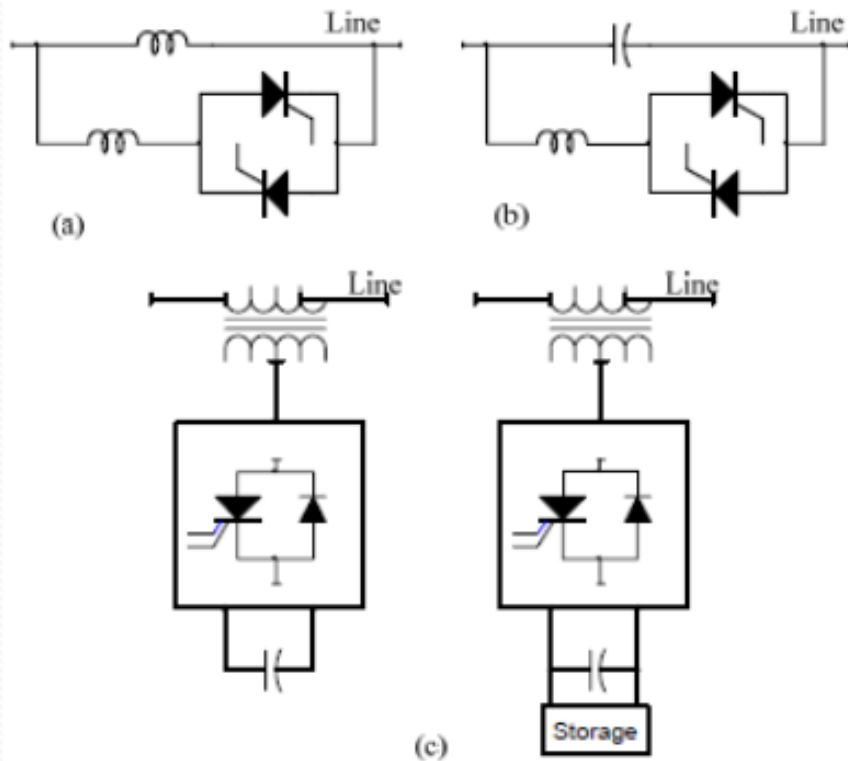
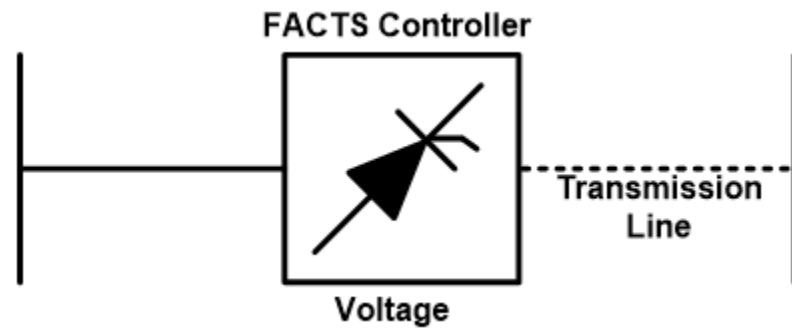


Figure. Series-connected FACTS controllers: (a) TCSR/TSSR (b) TSSC (c) SSSC [2]

[2] N. G. Hingorani, L. Gyugyi, 'Understanding FACTS, Concepts and Technology of Flexible AC Transmission Systems, IEEE Press 2000.

Thyristor Controlled Series Capacitor (TCSC)
 Thyristor Controlled Series Reactor (TCSR)
 Thyristor Switched Series Capacitor (TSSC)
 Static Synchronous Series Compensator (SSSC)



Increase in Power Transfer Capacity [series compensation]

The power transfer can be represented by the following equation:

$$P = \frac{V_S V_R}{X} \sin \delta$$

Now if the value of X can be reduced, the power transfer can be increased.

$$P = \frac{V_S V_R}{X_L - X_{CF}} \sin \delta$$

Where,

$X_L \longrightarrow$ Series inductive reactance of the line.

$X_{CF} \longrightarrow$ Controllable series capacitive reactance of the FACTS.

By controlling X_{CF} by means of bypass Thyristor switch the power transfer P is controlled.

FLEXIBLE AC TRANSMISSION SYSTEM (FACTS)

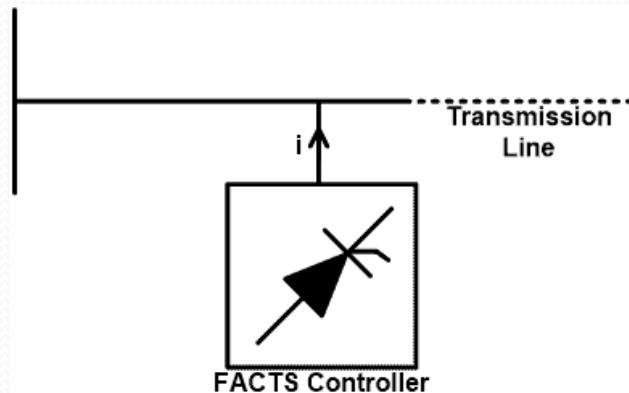
Shunt Connected Controller

This type of device is used to inject current to the power system at the point of connection. Similar to the series-connected controllers, it also consists of variable impedance like capacitor and inductor.

When a capacitor is used to connect parallel with the power system, the method is known as shunt capacitive compensation. When the transmission line consists of a more inductive load, it operates on a lagging power factor. This method is used to draw current leading to the source voltage to compensate for the lagging load with the help of a shunt connected capacitor.

When an inductor is used to connect in shunt with the power system, the method is known as shunt inductive compensation. Generally, this method is not more useful with the transmission network. But in case of a very large transmission line, the load is disconnected or it operates on no-load or less load condition, due to the Ferranti effect, the receiving end voltage increases then the sending end voltage. To avoid this condition, the shunt inductive compensator is used.

The example of this type of system is SVC (Static VAR compensator) and Static Synchronous Compensator).



Static VAR Compensator (SVC)

- Thyristor Controlled Reactor (TCR)
 - Thyristor Switched Capacitor (TSC)
 - Thyristor Switched Reactor (TSR)
- Static Synchronous Compensator (STATCOM)

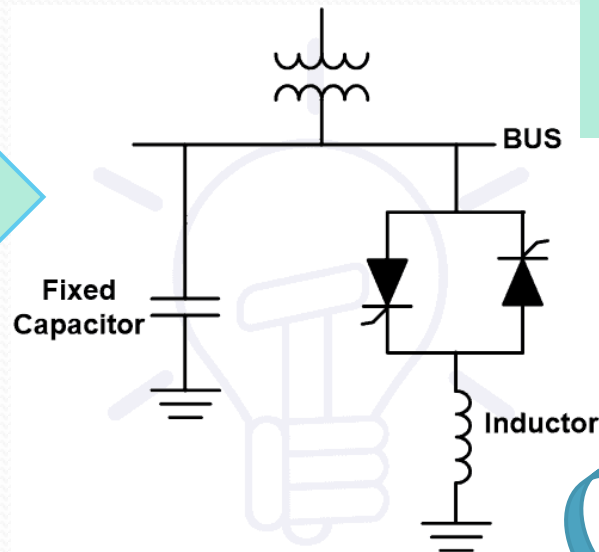
FLEXIBLE AC TRANSMISSION SYSTEM (FACTS)

Shunt Connected Controller

In **Static VAR Compensator**, a fixed capacitor bank is connected in parallel with the thyristor-controlled inductor. The firing angle of the thyristor controls the reactor and hence it controls the voltage across the inductor. Therefore, the power drawn by the inductor can be controlled. The basic diagram of the Static VAR Compensator is as shown in the below figure.

The SVC is used to improve power factor, regulating voltage, reduce harmonics, stabilize the transmission network. This type of device is also used in industries for reactive power control and improve power quality. The most commonly used SVCs are explained below.

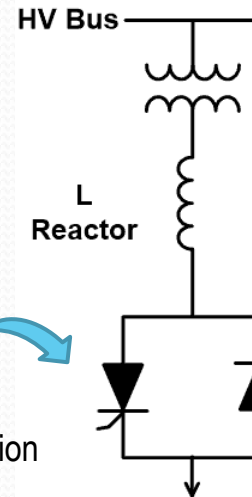
The basic diagram of the Static VAR Compensator



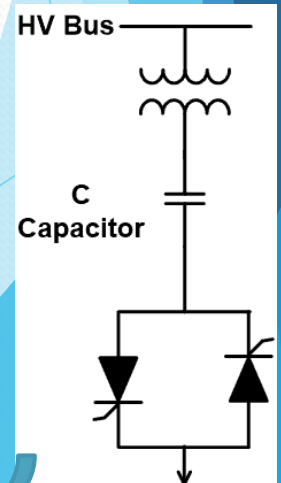
Used in light load or no-load condition

Static VAR Compensator (SVC)

- Thyristor Controlled Reactor (TCR)
- Thyristor Switched Capacitor (TSC)
- Thyristor Switched Reactor (TSR)



Used in heavy load condition



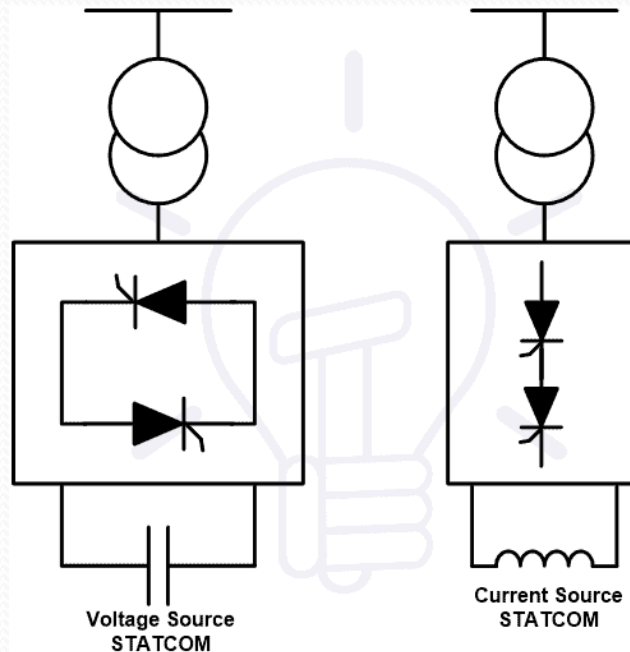
FLEXIBLE AC TRANSMISSION SYSTEM (FACTS)

Shunt Connected Controller

STATCOM is a power-electronics based voltage source converter that can regulate electricity in the transmission system by source or sink of reactive power. It also used to provide active power to the transmission line. Generally, the STATCOM is used in the transmission line which has very poor power factor and poor voltage regulation. It is the most commonly used device to improve the voltage stability of the power system.

It uses the charged capacitor as the input DC source and converts it into three-phase AC voltage as output by the voltage-controlled inverter. The output of the inverter is synchronized with the AC power of the power system. The inverter is connected in a shunt connection with the transmission line via a coupling transformer.

The output of STATCOM can be controlled by controlling the output of an inverter. The basic diagram of STATCOM is as shown in the below figure.

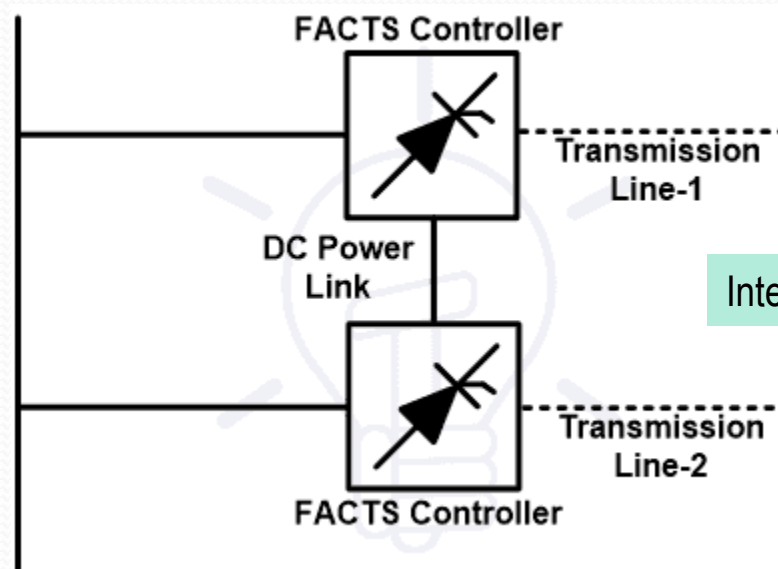


FLEXIBLE AC TRANSMISSION SYSTEM (FACTS)

Combined Series-Series Controller

In multiline transmission lines, a combination of separate series controllers is used in a coordinated manner to provide independent series reactive compensation for each line. But it can transfer real power with lines via power link.

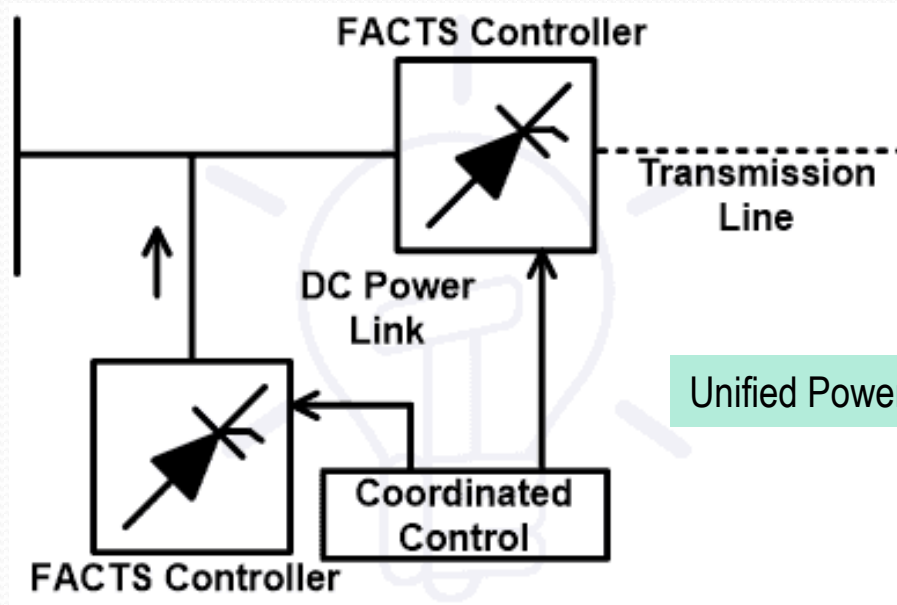
Or it can be connected with unified controllers that the DC terminals of converters are interlinked together. This will help to transfer the real power to the transmission line. An example of this type of system is the Interlink Power Flow Controller (IPFC).



FLEXIBLE AC TRANSMISSION SYSTEM (FACTS)

Combined Shunt-Series Controller

This type of controller is used to introduce voltage in parallel using the shunt controller and along with it used to introduce the current in series using the series controller. But both controllers must operate with coordination. The example of this type of controller is the Unified Power Flow Controller (UPFC).



Unified Power Flow Controller (UPFC)

FLEXIBLE AC TRANSMISSION SYSTEM (FACTS)

Advantages of FACTS

The advantages of FACTS devices are listed below:

- ☐ The FACTS devices are used to increase the power transfer capability of the transmission line. Hence, it saves costs to develop an entirely new transmission line.
- ☐ It increases the loading capability of the transmission line of their thermal capability.
- ☐ It reduces the amount of reactive power in the transmission line. In this way, the transmission line can transfer more amount of active power to the load.
- ☐ The FACTS devices are used to control the amount of flow of power through the transmission line. So, it helps to follow the guideline and contract of utility and ensure the optimum flow of power.
- ☐ It reduces the cost of power as it reduces the transmission cost.
- ☐ It increases the quality of power, voltage stability, thermal stability, and transient stability of the system.
- ☐ FACTS devices are not containing any environmentally hazardous waste material. Hence, this technology of environmentally friendly.
- ☐ It improves the power factor of the power system network.
- ☐ It increases the reliability and flexibility of the transmission network.

FLEXIBLE AC TRANSMISSION SYSTEM (FACTS)

Disadvantages of FACTS

The disadvantages of FACTS devices are as listed below:

- ❑ The FACTS devices are used power electronics switches to control supply or absorb the power. The major disadvantage of the use of power electronics switch is that it induces harmonics in the output signals. These harmonics enter into the power system network. So, the active filters are used to remove the harmonics.
- ❑ The initial cost of FACTS devices is very high.
- ❑ It needs continuous maintenance and the repair cost of these devices are very high.
- ❑ It uses converters. Hence, the amount of power loss occurs in the switches.
- ❑ If the current from the FACTS devices is increased due to any reason, there may be a chance to damage the power electronics switches used in the converter. And the cost of these switches is very high.

Reference:

Principles of Power System by V.K. Mehta and Rohit Mehta

Modern Power System Analysis, D P Kothari, I J Nagrath



QUESTIONS AND DISCUSSIONS

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